

a. Detailed L_2 -Convergence Check

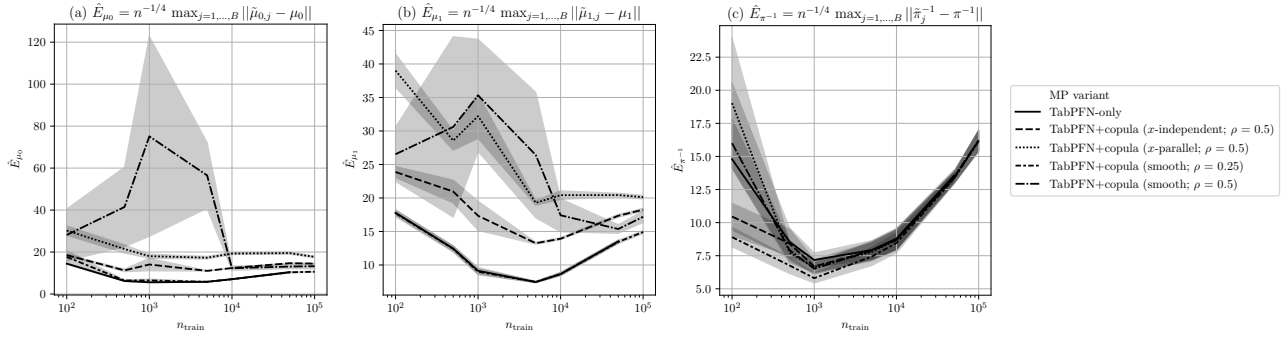


Figure 1. L_2 -convergence check for every nuisance function based on the synthetic data with varying size of the train data, n_{train} (here $d_x = 25$). Reported: mean L_2 error $\hat{E}_\diamond \pm \text{se}$ over 10 runs (lower is better), where $\diamond$ corresponds to different nuisance functions: (a) $\diamond = \mu_0$, (b) $\diamond = \mu_1$, and (c) $\diamond = \pi^{-1}$. **Results.** The performance improves with sample size up to a moderate regime, consistent with the L_2 -convergence condition required for the BvM theorem. However, beyond this regime, the concentration of the inverse propensity score posterior deteriorates, which reflects a difficulty of accurately recovering inverse propensity scores for propensities close to zero and one with TabPFN. Together, this confirms that, in practice, the contraction rates are sufficient to yield stable and well-aligned uncertainty estimates, yet only for medium-sized datasets.

b. Synthetic Dataset Ablations

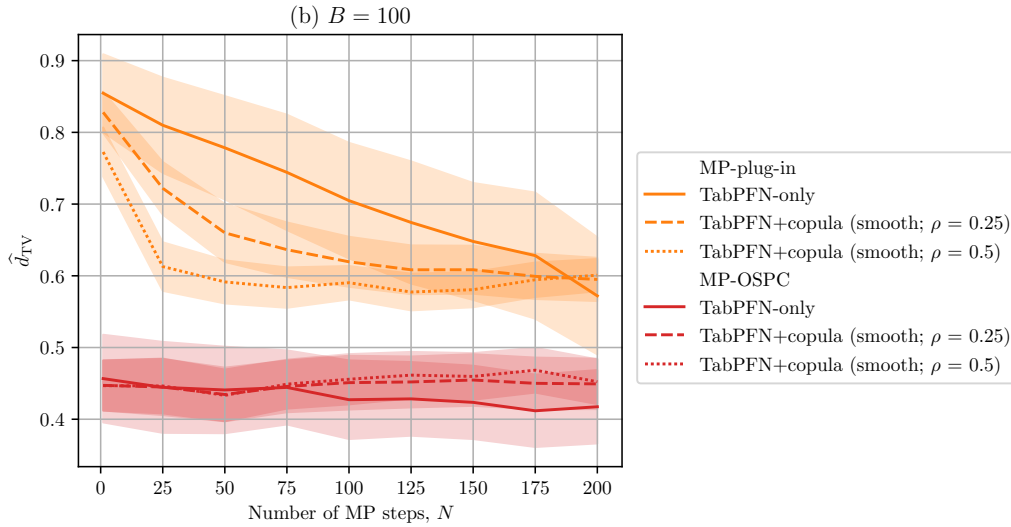


Figure 2. Quality of the asymptotic uncertainty for Bayesian ATE estimators based on the synthetic data with varying number of MP steps, N ($n_{\text{train}} = 500$, $d_x = 25$). Reported: mean $\hat{d}_{\text{TV}} \pm \text{se}$ over 40 runs (lower is better). **Results.** Our MP-based ATE posteriors converge already when $N = 100$, suggesting that the main source of error is the PPD of an underlying PFN.